Настройка crontab и автоматизация задач в Linux

Цель работы

Познакомиться с планировщиком задач cron, научиться настраивать автоматические задания через crontab, создавать, редактировать и удалять задания, а также анализировать их выполнение.

Практические задания

Задание 1: Создание простого cron-задания

Добавьте задание, которое будет записывать текущее время в файл каждые 5 минут:

```
dasha@dasha-VMware-Virtual-Platform:~$ crontab -e
crontab: installing new crontab
dasha@dasha-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
*/5 * * * * echo "Current time: $(date)" >> ~/cron_log.txt
dasha@dasha-VMware-Virtual-Platform:~$ cat cron_log.txt
Current time: Fri Apr 24 04:00:01 PM MSK 2026
dasha@dasha-VMware-Virtual-Platform:~$
```

```
dasha@dasha-VMware-Virtual-Platform:~$ cat cron_log.txt
Current time: Fri Apr 24 04:00:01 PM MSK 2026
Current time: Fri Apr 24 04:05:01 PM MSK 2026
Current time: Fri Apr 24 04:10:01 PM MSK 2026
Current time: Fri Apr 24 04:15:01 PM MSK 2026
Current time: Fri Apr 24 04:20:01 PM MSK 2026
```

Задание 2: Ежедневная резервная копия

Напишите простой скрипт backup.sh, который будет копировать содержимое папки ~/Documents в папку ~/Backup. Сделайте скрипт исполняемым и добавьте его в crontab для ежедневного выполнения в полночь:

```
GNU nano 7.2
#!/bin/bash
mkdir -p ~/Backup
cp -r ~/Documents/* ~/Backup/
```

```
dasha@dasha-VMware-Virtual-Platform:~$ nano backup.sh
dasha@dasha-VMware-Virtual-Platform:~$ chmod +x backup.sh
dasha@dasha-VMware-Virtual-Platform:~$ mkdir -p ~/Documents
dasha@dasha-VMware-Virtual-Platform:~$ echo "Test file content" > ~/Documents/test.txt
dasha@dasha-VMware-Virtual-Platform:~$ ./backup.sh
dasha@dasha-VMware-Virtual-Platform:~$ ls Backup
papka  test  test.txt
```

```
dasha@dasha-VMware-Virtual-Platform:~$ crontab -e
crontab: installing new crontab
dasha@dasha-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 * * * ~/backup.sh
dasha@dasha-VMware-Virtual-Platform:~$
```

```
0 0 * * * ~/backup.sh
dasha@dasha-VMware-Virtual-Platform:~$ ls -l Backup
total 12
drwx------ 2 dasha dasha 4096 Apr 24 16:16 papka
drwxrwxr-x 5 dasha dasha 4096 Apr 24 16:16 test
-rw-rw-r-- 1 dasha dasha   18 Apr 28 00:00 test.txt
dasha@dasha-VMware-Virtual-Platform:~$ grep CRON /var/log/syslog |grep backup |tail -5
2026-04-28T00:00:02.146167+03:00 dasha-VMware-Virtual-Platform CRON[21266]: (dasha) CMD (~/backup.sh)
```

Задание 3: Оповещение о дисковом пространстве

Создайте скрипт check_space.sh, который проверяет, есть ли на диске меньше 10% свободного места, и отправляет уведомление. Сделайте скрипт исполняемым и настройте crontab на ежедневный запуск скрипта в 8 утра:

```
  GNU nano 7.2                                  check_space.sh *
#!/bin/bash
threshold=10
current_usage=$(df / | grep / | awk '{print $5}' | sed 's/%//')

if [ "$current_usage" -ge "$threshold" ]; then
        echo "Warning: less than 10% free disk space" | mail -s "Disk space notification" dasha
fi
```

```
dasha@dasha-VMware-Virtual-Platform:~$ mail
"/var/mail/dasha": 1 message 1 unread
>U   1 Dasha              Fri Apr 24 16:44  16/669   Disk space notification
? 1
Return-Path: <dasha@dasha-VMware-Virtual-Platform>
X-Original-To: dasha
Delivered-To: dasha@dasha-VMware-Virtual-Platform.localdomain
Received: by dasha-VMware-Virtual-Platform.localdomain (Postfix, from userid 1000)
        id AAB92120E02; Fri, 24 Apr 2026 16:44:36 +0300 (MSK)
Subject: Disk space notification
To: dasha@dasha-VMware-Virtual-Platform.localdomain
User-Agent: mail (GNU Mailutils 3.17)
Date: Fri, 24 Apr 2026 16:44:36 +0300
Message-Id: <20260424134436.AAB92120E02@dasha-VMware-Virtual-Platform.localdomain>
From: Dasha <dasha@dasha-VMware-Virtual-Platform>
X-IMAPbase: 1777038279                    3
X-UID: 2
Status: O

Warning: less than 10% free disk space
? q
Saved 1 message in /home/dasha/mbox
Held 0 messages in /var/mail/dasha
dasha@dasha-VMware-Virtual-Platform:~$
```

```
0 8 * * * ~/check_space.sh
dasha@dasha-VMware-Virtual-Platform:~$ grep CRON /var/log/syslog | grep check_space | tail -5
2026-04-27T08:00:01.594161+03:00 dasha-VMware-Virtual-Platform CRON[20696]: (dasha) CMD (~/check_space.sh)
dasha@dasha-VMware-Virtual-Platform:~$
```

```
dasha@dasha-VMware-Virtual-Platform:~$ mail
"/var/mail/dasha": 6 messages 6 new
>N  1 Cron Daemon      Sun Apr 26 02:00  38/2119  Cron <dasha@dasha-VMware-Virtual-Platform> ~/clean_temp.sh
 N  2 Cron Daemon      Mon Apr 27 07:05  18/848   Cron <dasha@dasha-VMware-Virtual-Platform> ~/weekday_test.sh
 N  3 Cron Daemon      Mon Apr 27 07:10  18/848   Cron <dasha@dasha-VMware-Virtual-Platform> ~/weekday_test.sh
 N  4 Cron Daemon      Mon Apr 27 07:11  18/848   Cron <dasha@dasha-VMware-Virtual-Platform> ~/weekday_test.sh
 N  5 Cron Daemon      Mon Apr 27 07:12  18/848   Cron <dasha@dasha-VMware-Virtual-Platform> ~/weekday_test.sh
 N  6 Dasha            Mon Apr 27 08:00  13/606   Disk space notification
?
```

Задание 4: Очистка временных файлов

Создайте скрипт clean_temp.sh, который будет удалять временные файлы, старше 7 дней, из директории /tmp. Сделайте скрипт исполняемым и добавьте задание в crontab, чтобы скрипт выполнялся каждую неделю по воскресеньям в 2 часа ночи:

```
  GNU nano 7.2                                        clean_temp.sh *
#!/bin/bash
find /tmp -type f -mtime +7 -exec rm -f {} \;
```

```
0 2 * * 7 ~/clean_temp.sh
```

```
dasha@dasha-VMware-Virtual-Platform:~$ grep CRON /var/log/syslog | grep clean_temp | tail -5
2026-04-26T02:00:19.962301+03:00 dasha-VMware-Virtual-Platform CRON[20053]: (dasha) CMD (~/clean_temp.sh)
```

Задание 5: создать задание только по будням

```
  GNU nano 7.2                                        weekday_test.sh *
#!/bin/bash
echo "Weekday task executed at: $(date)"
```

```
dasha@dasha-VMware-Virtual-Platform:~$ nano weekday_test.sh
dasha@dasha-VMware-Virtual-Platform:~$ chmod +x weekday_test.sh
dasha@dasha-VMware-Virtual-Platform:~$ crontab -e
crontab: installing new crontab
dasha@dasha-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
```

```
* * * * 1-5 ~/weekday_test.sh >> ~/weekday.log 2>&1
dasha@dasha-VMware-Virtual-Platform:~$ cat weekday.log
Weekday task executed at: Mon Apr 27 07:13:01 AM MSK 2026
Weekday task executed at: Mon Apr 27 07:14:01 AM MSK 2026
Weekday task executed at: Mon Apr 27 07:15:01 AM MSK 2026
Weekday task executed at: Mon Apr 27 07:16:01 AM MSK 2026
```

Задание 6: настроить логирование выполнения

```
GNU nano 7.2                                    logged_task.sh *
#!/bin/bash
echo "Task started at: $(date)"
echo "Doing some work..."
sleep 2
echo "Task finished at $(date)"
echo "----------------------------"
```

```
dasha@dasha-VMware-Virtual-Platform:~$ nano logged_task.sh
dasha@dasha-VMware-Virtual-Platform:~$ chmod +x logged_task.sh
dasha@dasha-VMware-Virtual-Platform:~$ crontab -e
```

```
*/10 * * * * ~/logged_task.sh >> ~/task.log 2>&1
dasha@dasha-VMware-Virtual-Platform:~$ cat ~/task.log
Task started at: Sun Apr 26 11:00:01 AM MSK 2026
Doing some work...
Task finished at Sun Apr 26 11:00:03 AM MSK 2026
----------------------------
Task started at: Sun Apr 26 11:10:01 AM MSK 2026
Doing some work...
Task finished at Sun Apr 26 11:10:03 AM MSK 2026
----------------------------
dasha@dasha-VMware-Virtual-Platform:~$
```

Задание 7: Архивация с датой

```
GNU nano 7.2                                  /home/dasha/dated_archive.sh *
#!/bin/bash
DATE=$(date +%Y-%m-%d_%H-%M-%S)
tar -czf ~/archive_$DATE.tar.gz ~/Documents
echo "Archive created: archive_$DATE.tar.gz at $(date)"
```

```
dasha@dasha-VMware-Virtual-Platform:~$ chmod +x dated_archive.sh
dasha@dasha-VMware-Virtual-Platform:~$ ./dated_archive.sh
tar: Removing leading `/' from member names
Archive created: archive_2026-04-26_11-18-20.tar.gz at Sun Apr 26 11:18:20 AM MSK 2026
dasha@dasha-VMware-Virtual-Platform:~$ ls ./archive_*.tar.gz
./archive_2026-04-26_11-18-20.tar.gz
dasha@dasha-VMware-Virtual-Platform:~$
```

```
23 11 * * * ~/dated_archive.sh >> ~/archive.log 2>&1
dasha@dasha-VMware-Virtual-Platform:~$ cat archive.log
tar: Removing leading `/' from member names
Archive created: archive_2026-04-26_11-23-01.tar.gz at Sun Apr 26 11:23:01 AM MSK 2026
dasha@dasha-VMware-Virtual-Platform:~$
```

Задание 8: Проверка доступности сайта

```
dasha@dasha-VMware-Virtual-Platform:~/bash_practice$ cat check_site.sh
#!/bin/bash
SITE=$1
LOG_FILE="site_check.log"
ping -c 1 -W 2 "$SITE"

if [ $? -eq 0 ]; then
        echo "The site is available"
else
        echo "$(date '+%Y-%m-%d %H:%M:%S') - the site $SITE is unavailable" >> "$LOG_FILE"
        echo "The site is unavailable (the entry was added to $LOG_FILE)"
fi
dasha@dasha-VMware-Virtual-Platform:~/bash_practice$
```

```
0 12 * * * ~/bash_practice//check_site.sh google.com >> ~/bash_practice/site_check_cron.log 2>&1
dasha@dasha-VMware-Virtual-Platform:~/bash_practice$ cat site_check_cron.log
PING google.com (64.233.161.101) 56(84) bytes of data.
64 bytes from lh-in-f101.1e100.net (64.233.161.101): icmp_seq=1 ttl=128 time=21.7 ms

--- google.com ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 21.667/21.667/21.667/0.000 ms
The site is available
dasha@dasha-VMware-Virtual-Platform:~/bash_practice$
```